RasterMan

High-performance Manhattan Plot Visualization

Version 1.0.0 · GPL ≥ 3 · © 2026 Wentao Cai (caiwentao@caas.cn)
https://iasbreeding.cn/rasterman/

Overview

RasterMan uses adaptive C++ rasterization to render Manhattan plots of million-scale GWAS / eQTL datasets with ~20× speedup and ~87% memory reduction compared to standard ggplot2. Three usage modes are provided:

Mode	When to use
Cloud platform	Quick exploratory plots — no R installation needed
Rscript CLI	Batch processing, HPC pipelines, automation
R API	Integration into your own R scripts and workflows

Installation

Install from the source tarball (requires a C++ compiler and Rcpp):

```
install.packages("RasterMan_1.0.0.tar.gz", repos = NULL, type = "source")
```

Required dependencies (installed automatically):

```
install.packages(c("Rcpp", "ggplot2", "data.table", "dplyr", "magrittr",
"ggrepel"))
# Recommended for combined plots:
install.packages("patchwork")
```

Mode 1 — Cloud Platform

URL: https://iasbreeding.cn/rasterman/
No installation required. Upload your summary statistics file and configure the plot in the browser.

Workflow

① Upload file → ② Set column names → ③ Configure style → ④ Download plot

Supported Input Formats

Format	Notes
Plain text (.txt, .tsv, .csv)	Space / tab / comma delimited
Gzip compressed (.gz)	Recommended for large files
Maximum file size	500 MB

Step-by-Step

① Upload file

Click Upload and select your summary statistics file. The platform auto-detects column names; confirm or correct them in the column mapping panel.

② Column mapping

Field	Description	Common values
Chromosome	Integer chromosome identifier	Chr, CHR, CHROM
Position	Base-pair position	Pos, BP, POS
P-value	Association p-value	P, Pvalue, p_wald
Gene (cis/trans)	Gene name or ID	Gene, gene_name
Gene Chr/Start/End	Gene genomic coordinates (trans)	gene_chr, gene_start, gene_end
Effect (trans, opt.)	Beta / effect size for colour	beta, slope

③ Style options

Option	Values
Plot type	GWAS Manhattan · Cis-eQTL · Trans-eQTL
Color scheme	modern · classic · colorblind · pastel · vibrant · gray
Significance threshold	5e-8 (GWAS default) | 1e-5 (eQTL default)
Lead SNP annotation	Enable/disable, window size
Gene annotation	Upload BED file
Rasterization method	auto · smart · multilevel · adaptive · basic · off
Output format	PNG · TIFF · PDF

Option	Values
Resolution (DPI)	150 / 300 / 600

④ Download

Click Generate Plot → preview the result → click Download.

Mode 2 — Rscript Command Line

Setup

```
# Get the path to installed CLI scripts
SCRIPTS=$(Rscript -e "cat(system.file('scripts', package='RasterMan'))")

# Optional: copy to your project directory
cp $SCRIPTS/plot_gwas.R        ./
cp $SCRIPTS/plot_cis_eqtl.R   ./
cp $SCRIPTS/plot_trans_eqtl.R ./
```

GWAS Manhattan Plot

```
Rscript $SCRIPTS/plot_gwas.R <input_file> [options]
Rscript $SCRIPTS/plot_gwas.R --help
```

Examples

GCTA .mlma:

```
Rscript $SCRIPTS/plot_gwas.R results.mlma \
    --chr-col Chr --bp-col bp --p-col p
```

PLINK .assoc — lead SNPs + combined QQ:

```
Rscript $SCRIPTS/plot_gwas.R plink.assoc \
    --chr-col CHR --bp-col BP --p-col P --snp-col SNP \
    --anno-snp --qq-combine --color-scheme colorblind --dpi 300
```

BOLT-LMM — gene annotation, high-res TIFF:

```
Rscript $SCRIPTS/plot_gwas.R bolt.results \
    --chr-col SNP_CHR --bp-col SNP_BP --p-col P_BOLT_LMM \
    --anno-snp --anno-gene --gene-bed hg38_genes.bed \
    --format tiff --dpi 600 --width 14 --height 6
```

All GWAS Options

Option	Default	Description
--chr-col NAME	CHR	Chromosome column
--bp-col NAME	BP	Position column
--p-col NAME	P	P-value column
--snp-col NAME	SNP	SNP ID column
--sig-threshold VALUE	5e-8	Genome-wide significance line
--anno-snp	off	Annotate lead SNPs

Option	Default	Description
--window-size BP	500000	Clumping window size (bp)
--max-anno N	50	Max SNP labels
--anno-gene	off	Annotate nearest gene
--gene-bed FILE	—	Gene BED file (chr, start, end, name)
--gene-distance BP	50000	Max gene–SNP distance
--qq-plot	off	Save QQ plot separately
--qq-only	off	QQ plot only
--qq-combine	off	Manhattan + QQ in one figure
--color-scheme NAME	modern	Color scheme (see Color Schemes table)
--color-even HEX	—	Custom even-chromosome color
--color-odd HEX	—	Custom odd-chromosome color
--raster-method METHOD	auto	Rasterization method
--format FORMAT	png	Output format (png/tiff/pdf)
--dpi N	300	Resolution
--width INCHES	12	Plot width in inches
--height INCHES	6	Plot height in inches
--output-prefix PREFIX	gwas_plot	Output filename prefix
--no-background	off	Remove chromosome background bands

Cis-eQTL Manhattan Plot

```
Rscript $SCRIPTS/plot_cis_eqtl.R <input_file> [options]
Rscript $SCRIPTS/plot_cis_eqtl.R --help
```

Examples

Basic usage:

```
Rscript $SCRIPTS/plot_cis_eqtl.R cis.txt.gz \
    --chr-col Chr --bp-col Pos --p-col P --gene-col gene_name
```

Large dataset with highlighted genes:

```
Rscript $SCRIPTS/plot_cis_eqtl.R big_cis.txt.gz \
    --chr-col Chr --bp-col Pos --p-col P --gene-col gene_name \
    --raster-method multilevel \
    --anno-gene sig_genes.txt --highlight-color "#E74C3C" \
    --color-scheme colorblind --dpi 300
```

All Cis-eQTL Options

Option	Default	Description
--chr-col NAME	Chr	Chromosome column
--bp-col NAME	Position	Position column

Option	Default	Description
--p-col NAME	Pvalue	P-value column
--gene-col NAME	Gene	Gene name column
--snp-col NAME	—	SNP ID column (auto chr:pos if absent)
--sig-threshold VALUE	1e-5	Significance threshold
--min-gene-logp VALUE	5	Min -log10(P) for gene labels
--max-labels N	all	Max gene labels
--anno-gene FILE	—	cis-eGene list to highlight (one per line)
--highlight-color COLOR	red	Highlight point color
--highlight-size SIZE	1.5	Highlight point size
--color-scheme NAME	modern	Color scheme
--color-even HEX	—	Custom even-chromosome color
--color-odd HEX	—	Custom odd-chromosome color
--raster-method METHOD	auto	Rasterization method
--output FILE	cis_eqtl_manhattan.png	Output filename
--width INCHES	14	Plot width
--height INCHES	7	Plot height
--dpi N	300	Resolution
--no-background	off	Remove chromosome bands

Trans-eQTL 2D Manhattan Plot

```
Rscript $SCRIPTS/plot_trans_eqtl.R <input_file> [options]
Rscript $SCRIPTS/plot_trans_eqtl.R --help
```

Examples

Basic (color by -log10P):

```
Rscript $SCRIPTS/plot_trans_eqtl.R trans.txt.gz \
    --chr-col Chr --pos-col Pos --p-col P \
    --gene-chr-col gene_chr --gene-start-col gene_start --gene-end-col gene_end
```

Color by effect direction (--slope-col beta):

```
Rscript $SCRIPTS/plot_trans_eqtl.R trans.txt.gz \
    --chr-col Chr --pos-col Pos --p-col P \
    --gene-chr-col gene_chr --gene-start-col gene_start --gene-end-col gene_end \
    --slope-col beta --color-scheme modern --dpi 300
```

All Trans-eQTL Options

Option	Default	Description
--chr-col NAME	Chr	SNP chromosome column
--pos-col NAME	Pos	SNP position column

Option	Default	Description
--p-col NAME	Pvalue	P-value column
--gene-chr-col NAME	Gene_chr	Gene chromosome column
--gene-start-col NAME	Start	Gene start position column
--gene-end-col NAME	End	Gene end position column
--snp-col NAME	—	SNP ID column (auto chr:pos if absent)
--gene-col NAME	—	Gene name column (optional)
--slope-col NAME	—	Beta/effect column for gradient coloring
--color-scheme NAME	modern	Gradient color scheme
--color-low HEX	—	Custom negative-effect color
--color-high HEX	—	Custom positive-effect color
--raster-method METHOD	auto	Rasterization method (basic/off)
--output FILE	trans_eqtl_manhattan.png	Output filename
--width INCHES	10	Plot width
--height INCHES	10	Plot height
--dpi N	300	Resolution
--no-background	off	Remove chromosome bands

Mode 3 — R API

GWAS

```
library(RasterMan)

gwas <- read_gwas("results.mlma",
                  chr_col = "Chr", bp_col = "bp", p_col = "p")

prep     <- prep_gwas(gwas)
gwas_df  <- prep$gwas
chr_info <- prep$chr_info

params <- list(
  sig_threshold     = 5e-8,
  color_scheme      = "modern",   # modern/classic/colorblind/pastel/vibrant/gray
  # color_even      = "#2E86AB", # override even-chromosome color
  # color_odd       = "#A23B72", # override odd-chromosome color
  anno_snp          = TRUE,
  window_size       = 500000,
  max_anno          = 50,
  anno_gene         = FALSE,
  raster_method     = "auto",
  output_format     = "png",
  mode              = "manhattan",
```

```
  adaptive_sampling = TRUE,
  anno_threshold    = 5e-8
)

p  <- plot_gwas(gwas_df, chr_info, params)
pq <- plot_qq(gwas_df, params)

ggplot2::ggsave("manhattan.png", p,  width = 14, height = 6, dpi = 300)
ggplot2::ggsave("qqplot.png",    pq, width = 6,  height = 6, dpi = 300)
```

Cis-eQTL

```
library(RasterMan)

cis <- read_cis("cis.txt.gz",
                chr_col  = "Chr",
                bp_col   = "Pos",
                p_col    = "P",
                gene_col = "gene_name")

prep     <- prep_cis(cis)
cis_df   <- prep$gwas
chr_info <- prep$chr_info

lead      <- lead_snps(cis_df, min_gene_logp = 5)
highlight <- readLines("sig_genes.txt")  # or NULL

params <- list(
  sig_threshold   = 1e-5,
  color_scheme    = "colorblind",
  # color_even    = "#0173B2",
  # color_odd     = "#DE8F05",
  highlight_color = "#E74C3C",
  highlight_size  = 1.5,
  point_size      = 0.8,
  label_size      = 3,
  max_labels      = 50,
  chr_background  = TRUE,
  raster_method   = "auto",
  raster_width    = 2000L,
  raster_height   = 1000L
)

p <- plot_cis(cis_df, chr_info, params, lead, highlight)
ggplot2::ggsave("cis_manhattan.png", p, width = 14, height = 7, dpi = 300)
```

Trans-eQTL

```
library(RasterMan)

prep <- prep_trans(
  read_trans("trans.txt.gz"),
  chr_col        = "Chr",
  pos_col        = "Pos",
```

```
  p_col          = "P",
  gene_chr_col   = "gene_chr",
  gene_start_col = "gene_start",
  gene_end_col   = "gene_end",
  slope_col      = "beta"       # NULL -> colour by -log10(P)
)

params <- list(
  color_scheme  = "modern",
  # color_low   = "#2E86AB",
  # color_high  = "#A23B72",
  raster_method = "auto",
  raster_width  = 2000L,
  title         = "Trans-eQTL Manhattan"
)

p <- plot_trans(prep, prep$snp_chr_info, prep$gene_chr_info, params)
ggplot2::ggsave("trans_manhattan.png", p, width = 10, height = 10, dpi = 300)
```

Color Schemes

GWAS and cis-eQTL use alternating odd/even chromosome colors. Trans-eQTL uses a diverging gradient for effect direction. Specify with --color-scheme (CLI) or color_scheme (R API). Override individual colors with --color-even / --color-odd or color_even / color_odd.

Scheme	Odd chr / Negative effect	Even chr / Positive effect
modern	#A23B72	#2E86AB
classic	#FFA500	#4169E1
colorblind	#DE8F05	#0173B2
pastel	#FFB347	#AEC6CF
vibrant	#FF6B6B	#4ECDC4
gray	#D3D3D3	#808080

Rasterization Methods

Method	Recommended for	Notes
auto	All datasets	Selects best method automatically (default)
smart	100k – 2M pts	Density-aware; preserves significant signals
multilevel	2M – 30M pts	Fine grid for signals + coarse background ★
adaptive	500k – 5M pts	Subdivides high-density regions
basic	Any size	Fastest; may obscure weak signals
off	< 200k pts	Standard ggplot2 vector rendering

Performance Reference — cis-eQTL, 31M data points

Method	Time	Quality	Memory
off (no raster)	~30 min	100%	overflow
basic	~2 min	85%	low
smart	~1.5 min	92%	medium
multilevel ★	~1 min	95%	medium
adaptive	~2.5 min	93%	high

★ *multilevel is recommended for datasets >2M points: all significant signals are preserved at full resolution while background points are aggressively compressed.*

Input Format Reference

Common column name mappings for major GWAS / eQTL software:

Software	--chr-col	--bp-col	--p-col	--snp-col
GCTA .mlma	Chr	bp	p	SNP
PLINK .assoc	CHR	BP	P	SNP
BOLT-LMM	SNP_CHR	SNP_BP	P_BOLT_LMM	SNP
GEMMA	chr	ps	p_wald	rs
SAIGE	CHR	POS	p.value	SNPID
REGENIE	CHROM	GENPOS	LOG10P *	ID

* REGENIE outputs -log10(P); read_gwas() auto-detects and converts.